

Deep Learning for Content Moderation on the Shareish Solidarity Platform

Progress Report — Follow-up to October 2025

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Since October — What Changed

October 2025 — Open question

Which training approach? Option A (safety-specialized) vs B (general base model)

Two-tier architecture: “possible approach under evaluation”

Evaluation framework: designed, not yet tested

French support: “critical but underexplored”

LoRA fine-tuning on French data: planned

Alan GPU cluster: “to learn”

April 2026 — Status

Resolved. Option A — LlamaGuard-3-1B and ShieldGemma-2b evaluated as primary candidates

Confirmed. Detoxify-M (Tier 1) + LoRA LG-1B (Tier 2), measured end-to-end

Implemented. 10 models $\times$ 8 datasets; extended with a **deployability axis** (VRAM, latency, energy)

Confirmed empirically. EN $\rightarrow$ FR F1 gap up to -0.686 (KoalaAI); French is a hard constraint

Executed — partially. FR-Hate Superset adapters increase precision (TNR $\uparrow$); Reddit-FR evaluation pending; domain closeness identified as a key variable for recall

Operational. Used for 48 h baseline run + LoRA training jobs

Context: Shareish & the Content Moderation Problem

Shareish — open-source platform for solidarity practices
mutual aid, generalized exchange, gift economy (ULiège)

- Everything is free — no real or virtual currency
- Request or offer material, manual, intellectual, or logistical aid
- French-speaking community (Belgium/France)
- No dedicated moderation infrastructure

user-reported harmful content goes unfiltered

Problem: Deploy an automated content moderator that is
both *accurate on French text* and *feasible on NGO hardware*.

Thesis scope

- Evaluate 10 pre-trained models on 8 datasets
- Primary axis: **French F1**
- Secondary axis: **Deployability**
- Fine-tune promising candidates (LoRA)
- Propose a production architecture

The Deployability Axis — Why It Matters Here

Oct 2025: evaluation framework covered Technical, Legitimacy, Operational, Robustness, Impact — but VRAM and inference speed were not explicit hard constraints. This emerged as the central finding of Phase 1.

Deployability metrics collected:

- Peak GPU memory (MB) at inference
- Average inference latency (ms/sample)
- Energy consumption (kWh / run)
- CO₂ equivalent (kg)

Novel evaluation angle

Most NLP benchmarks rank models by accuracy alone.

A model requiring 24 GB VRAM does not exist as a deployment option for a small NGO.

This thesis treats VRAM and latency as *hard constraints*, not afterthoughts.

Hardware tiers defined:

- **CPU-only** — zero GPU requirement
- **≤6 GB VRAM** — consumer GPU (RTX 3060)
- **6–16 GB** — prosumer GPU
- **>16 GB** — non-deployable for Shareish

Proposed Evaluation Framework

Primary development metric: F1 (macro)

Consistent with 88% of the literature; simple to compare across models and datasets.

Imbalance correction: MCC

Reddit-FR is class-imbalanced. F1 at a fixed threshold can be gamed by predicting the majority class. MCC accounts for all four cells of the confusion matrix without requiring a threshold sweep.

Secondary (threshold-independent): AUC-PR

AUC-ROC is misleading on imbalanced sets — FPR stays low when negatives are abundant. AUC-PR degrades visibly when the model makes false-positive errors, which is the critical failure mode for content moderation.

Reporting framework (5 dimensions):

1. **Technical:** F1, MCC, AUC-PR per dataset
2. **Robustness:** HateCheck pass rate (target $\geq 25/29$)
3. **Operational:** VRAM, latency (ms/sample), energy
4. **Legitimacy:** FPR disparity across groups, consistency
5. **Impact:** Human–AI agreement on Shareish data (Phase 3)

Phase 1 — Evaluation Setup

Oct 2025: “Option A (safety-specialized) vs B (general base model)?” → Answered: we evaluated both tiers of Option A across 10 models.

10 evaluated models:

Model	Tier
Detoxify-multilingual	CPU
Detoxify-unbiased	CPU
KoalaAI	CPU
EthicalEye	CPU
CitizenLab	CPU
LlamaGuard-3-1B	3 GB GPU
ShieldGemma-2b	5.7 GB GPU
LlamaGuard-3-8B	16 GB GPU
ShieldGemma-9b	18 GB GPU
Mistral-7B	14 GB GPU

8 datasets:

- **HateCheck-EN/FR** — functionality tests (counter-speech, slurs. . .)
- **FR-Hate Superset** — French hate speech corpus
- **ToxiGen** — LLM-generated implicit toxicity
- **OpenAI Moderation** — multi-category EN
- **Civil Comments** — Wikipedia talk-page toxicity
- **Reddit-EN / Reddit-FR** — social media, balanced

Protocol: threshold 0.5 · **Primary metric:** HC-FR F1

Phase 1 — Key Findings

What works:

- ✓ **Detoxify-M**: CPU, 6 ms, HC-FR 0.787, robust counter-speech
- ✓ **LG-1B**: 3 GB, 33 ms, HC-FR 0.674 — improvable via fine-tuning
- ✓ **SG-2b**: best raw FR F1 (0.858) but counter-speech flaw

What does not work:

- ✗ KoalaAI, Detoxify-unbiased: HC-FR F1 ≈ 0.10
- ✗ All >6 GB models: ruled out by hardware constraint

Methodological findings:

- EN performance is not a reliable proxy for FR (gap up to -0.686)
- Inference bugs fixed in v1 $\rightarrow$ v3: ShieldGemma wrong generation method; KoalaAI argmax vs. sum of unsafe probabilities
- Checkpointing essential: full run took ~ 48 h on Alan A5000

Conclusion of Phase 1

Two candidates selected for fine-tuning:

LG-1B (primary) and **SG-2b** (secondary)

Phase 1 — French Performance Overview

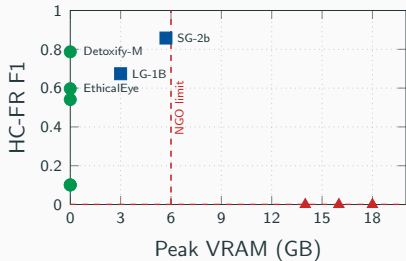
Model	HC-FR	RD-FR	FR-Hate	VRAM	ms/samp.
ShieldGemma-2b	0.858	0.311	0.648	5.7 GB	24
Detoxify-M	0.787	0.547	0.634	CPU	6
LlamaGuard-3-1B	0.674	0.398	0.372	3.0 GB	33
EthicalEye	0.597	— ^a	— ^a	CPU	— ^a
CitizenLab	0.541	— ^a	— ^a	CPU	— ^a
KoalaAI	0.101	— ^a	— ^a	CPU	8
Detoxify-unbiased	0.101	— ^a	— ^a	CPU	4
LlamaGuard-3-8B	— ^b	— ^b	— ^b	16 GB	—
ShieldGemma-9b	— ^b	— ^b	— ^b	18 GB	—
Mistral-7B	— ^b	— ^b	— ^b	14 GB	—

^a Omitted: model eliminated at HC-FR; further evaluation not prioritised.

^b Excluded by VRAM constraint (>6 GB); results exist but are not deployment-relevant.

EN→FR gap: KoalaAI HC-EN 0.787 vs HC-FR 0.101 ($\Delta = -0.686$) — EN performance is not a reliable proxy for French viability.

Phase 1 — Deployability vs. Accuracy Trade-off



Findings:

- Models >6 GB excluded by constraint alone
- **SG-2b** leads HC-FR (**0.858**, 5.7 GB, 24 ms)
- **Detoxify-M**: only competitive CPU option (**0.787**, 6 ms)
- Detoxify-M is 4× faster than LG-1B

Efficiency frontier

Detoxify-M and LG-1B sit on the Pareto frontier of French F1 × VRAM × latency.

Phase 1 — HateCheck Functionality Breakdown

HateCheck tests 29 functional categories (slurs, derogation, counter-speech, spelling variants. . .)

ShieldGemma-2b: counter-speech failure

Category	SG-2b	Detoxify-M
counter_quote_nh	0.054	0.921
counter_ref_nh	0.128	0.887
Hateful categories (avg)	0.91	0.74

Counter-speech quotes hateful language to rebut it.

SG-2b classifies the quoted slur as the signal — *correct-rate* 5.4%.

Deployment blocker: users reporting content by quoting it would themselves be flagged.

Why it matters:

- Aggregated F1 masks this failure (HC-FR F1 = 0.858 overall)
- Functionality-level breakdown is necessary for production safety
- Detoxify-M is robust to counter-speech despite lower overall F1

Lesson

A high aggregate score does not certify fitness for deployment. Model selection must include targeted stress tests.

Phase 2 — LoRA Fine-Tuning Setup

Oct 2025: “fine-tune on French datasets (10K–50K examples), then active learning with Shareish data” → executed as LoRA on FR-Hate Superset + Reddit-FR.

Why LoRA (Low-Rank Adaptation)?

- Full fine-tuning of LG-1B / SG-2b requires 16–40 GB VRAM
- LoRA injects trainable low-rank adapters into attention layers
- Reduces training VRAM to ~6–8 GB — fits A5000 (24 GB)
- Adapter size: ~10 MB vs. full checkpoint (~4 GB)

Configuration:

Rank r	16
Alpha α	32
Learning rate	2×10^{-4}
Epochs	3 (best = epoch 1)
Batch size	8
GPU	NVIDIA A5000 (24 GB)

Training data: FR-Hate Superset + Reddit-FR **Models:** LlamaGuard-3-1B, ShieldGemma-2b

Hypothesis: Joint training covers both formal and informal French hate speech styles.

Phase 2 — LoRA Fine-Tuning Experiments

Experiment 1 — Separate adapters on FR-Hate Superset

20% held-out test set, no overlap with training data.

Model		F1	Prec.	TPR	TNR	F1 improves for
LG-1B	Baseline	0.371	0.285	0.532	0.591	
	LoRA	0.557	0.640	0.493	0.915	
SG-2b	Baseline	0.413	0.405	0.420	0.811	
	LoRA	0.534	0.758	0.413	0.960	

both models.

Gain is **precision-driven** (TNR $\uparrow\uparrow$, TPR $\approx$ flat).

Models learn what is *not* harmful — not what *is*.

Experiment 2 — Reddit-FR adapter (pending)

- LG-1B adapter exists — fair eval not yet submitted
- SG-2b training failed (SLURM timeout) — needs resubmit with `--epochs 1`
- Contaminated run showed severe TPR collapse (0.298 $\rightarrow$ 0.089) — fair eval will clarify

Next experiment — Joint training

Train one adapter on FR-Hate + Reddit-FR simultaneously.

Hypothesis: combined signal may force generalisation across formal and informal registers.

Requires a proper held-out eval from the start.

Phase 2 — Domain Closeness & Data Collection

Why publicly available data is insufficient

Training corpora are *domain-distant* from real Shareish content:

- FR-Hate Superset: formal academic sentences, explicit slurs
- Reddit-FR: informal text, irony, noisy community labels
- Shareish: solidarity context, Belgian French, community-specific norms

Fine-tuning on formal hate corpora teaches models which *known patterns* are not harmful (precision $\uparrow$), but cannot generalise to unseen platform-specific content.

Hypothesis: data from platforms with similar distributions to Shareish should transfer better.

Data Collection — Phase 3

Shareish data would be ideal — but there is not enough available data.

Strategy: collect from solidarity and exchange platforms with similar content distributions:

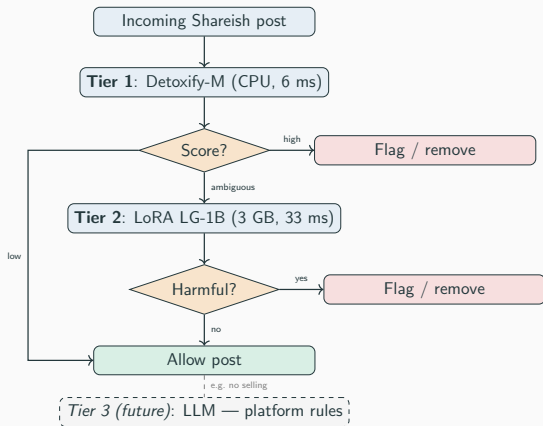
- Facebook solidarity groups
- Commercial exchange groups
- Vinted, donnons.org

Disclaimer — data sourcing

Scraping external platforms is **prohibited by Terms of Use** and raises **GDPR concerns**. Must use licensed corpora or explicit user consent.

Recommendation — Two-Tier Production Architecture

Oct 2025: "possible two-tier hybrid — Detoxify pre-filter + Llama Guard for complex cases (under evaluation)" → now a concrete recommendation backed by Phase 1 & 2 numbers.



CPU-only: Detoxify-M alone
HC-FR 0.787, 6 ms, zero GPU.

With 3–6 GB GPU: two-tier
Tier 1 routes on score: high → flag,
low → allow (never reaches Tier 2),
ambiguous → Tier 2.

Advantages:

- Counter-speech protected end-to-end
- Tier 2 only sees genuinely ambiguous content
- Combined latency < 39 ms

Phase 3 — Next Steps

Data Collection (primary focus):

1. **Source domain-close data:** collect from solidarity and exchange platforms (Facebook groups, donnons.org, Vinted...)
2. **Annotation:** label for harmful content with platform-specific patterns in mind
3. **Synthetic augmentation:** AI-generated examples matching platform register

Terms of Use and GDPR compliance must be addressed before any scraping.

Central hypothesis for Phase 3

Data from platforms with similar distributions to Shareish will improve recall where publicly available French corpora could not.

Fine-tuning & evaluation:

4. **Reddit-FR fair eval:** submit LG-1B adapter; resubmit SG-2b (`--epochs 1`)
5. **Joint training:** combined FR-Hate + Reddit-FR adapter with proper held-out eval
6. **End-to-end two-tier eval:** combined F1, latency, recall ceiling

Optional / advanced:

- **QLoRA:** reduce training VRAM to ~ 3 GB

Thank you for your attention!